



Unveiling route choice strategy heterogeneity from smart card data in a large-scale public transport network

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ABSTRACT

We contribute to the understanding of public transport passenger route choice behavior by developing and applying methods that capture behavioral strategies by making use of smart card data. We begin by proposing the classification of possible route choice behavioral strategies in two groups: disaggregated strategies and aggregated strategies. In the former, the alternatives correspond to itineraries, which are fixed sequences of stops and public transport lines. In the latter, common line alternatives are considered, which are combinations of itineraries defined under given criteria. Almost all route choice models use consideration sets composed only of itineraries, while public transport assignment models for strategic analysis mostly use a version of the common lines approach. We postulate that this dichotomy is inappropriate and that, instead, heterogeneity exists in the route choice strategy, both between users and across contexts. With the aim of verifying this hypothesis, we first propose an indicator function constructed as the difference between expected and observed trips for a given behavioral assumption. We apply then the indicator to a case study based on smart card data from the city of Santiago, Chile, from which we find evidence of heterogeneity. We identify individuals that follow either an aggregated or a disaggregated strategy, as well as others who seem to be using a combination of both strategies. We further analyze the heterogeneity hypothesis using an integrated discrete choice and latent class approach, which we apply to the same case study. This approach involves estimating path-size logit models built with alternatives from disaggregated and aggregated strategies, as well as a latent class model built from a combination of both. It also addresses methodological challenges related to the definition of the consideration set and the correction of endogeneity. Results confirm the heterogeneity hypothesis, suggesting that the mean probability that passengers belong to the class that uses a disaggregated strategy for route choice is 51.2%, and that this heterogeneity markedly occurs between individuals, not within their choices. We also find that the latent class considering aggregated alternatives appear to prefer bus over metro, while the latent class considering disaggregated alternatives prefer metro over bus. The fact that waiting time is relatively more burdensome for the aggregated strategy class is in line with this class's preference for common lines. Walking time and bus crowding are more burdensome for the disaggregated strategy class, in line with their observed modal preferences.

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1. Introduction

Modeling the route choices of passengers in public transport systems is a well-known complex problem, which is relevant to the improvement of public transport network planning, design, and assessment. In this area, two modeling approaches can be distinguished, depending on whether the analysis focuses on route choice behavior modeling, or public transport assignment models for strategic analysis.

Most studies on public transport passengers' route choice behavior have used the Multinomial Logit (MNL) discrete choice model (Grison et al., 2017; Guo, 2011; Jánošíková et al., 2014; Nassir et al., 2018; Raveau & Muñoz, 2014; Raveau et al., 2014; Raveau et al., 2011; Vrtic & Axhausen, 2002; Kim et al., 2020), with consideration sets mostly composed of itineraries, where all lines belonging to a set of relevant alternatives are regarded as different options. The use of the MNL model for this problem implies that the correlation due to overlapping route segments is ignored, as MNL model assumes independence of alternatives. To address this limitation, the analytical approach of Path Size (PS) Logit models have often been adopted, which accounts for the correlation by adding a deterministic term that reduces the utility of overlapped alternatives (Anderson et al., 2017; Bovy & Hoogendoorn-Lanser, 2005; De Grange et al., 2012; Hoogendoorn-Lanser et al., 2005; Nielsen et al., 2021; Tan et al., 2015; Yap et al., 2020). In this study, we call this approach the disaggregated strategy.

Transit assignment models make use of the concept of common lines and hyperpaths (Chriqui & Robillard, 1975; Cominetti & Correa, 2001; De Cea & Fernández, 1993; Nguyen & Pallottino, 1988; Spiess & Florian, 1989). The common lines problem was originally described by Chriqui & Robillard (1975) as the strategy of identifying a subset of public transport lines that minimizes the total expected travel time. According to this principle, passengers will take the first line of the common lines set that arrives at the bus stop. Chriqui & Robillard (1975) defined the common lines set in a portion of the route between two stops. Later, the definition of common lines is extended to a network with the name of shortest hyperpath by Nguyen & Pallottino (1988), where the first line that arrives to the stop defines the possible following characteristic of the route, e.g., the number of transfers. The common line and the hyperpath approaches have been successfully implemented for public transit assignment models for strategic analysis in software packages such as EMME/2 (INRO 1996) and ESTRAUS (De Cea et al., 2003). We call this approach aggregated strategy.

The dichotomy between the disaggregated and aggregated approaches to, seemingly, the same problem seems inappropriate. For example, one possible implication of the concept of common lines for the modeling of discrete choices is that common lines may be perceived as part of the same alternative for passengers that face a route choice decision. However, to the best of our knowledge, only two studies have explicitly considered common lines between consecutive stops as part of the same alternative in a discrete choice framework. The first is Raveau & Muñoz (2014), who developed a MNL model using common lines as part of the same route alternative, and the second is Schmöcker et al. (2013), who developed a bi-level route choice model using hyperpaths as the consideration set. On the other hand, Kim et al. (2020) and Tan et al. (2015) use a similar idea that considers that overlapping sections between two consecutive stops of lines are regarded as one alternative. Their approach captures partially the common lines principle as it does not include those common lines that do not overlap their routes.

There is no theoretical or empirical justification for not using common lines in route choice behavior modeling. Raveau & Muñoz (2014) asked a sample of public transport users to self-report the route choice strategies that they use, finding that some passengers stated choosing between itineraries, but others declared using common line or hyperpath strategies. In this study, we confirm and deepen the empirical findings by Raveau & Muñoz (2014) by using a novel methodology that makes use of massive revealed preference data gathered from smart cards (SC), circumventing the problem of response bias that is inherent in self-reported data. The analysis is performed by estimating three types of models: i) a PS logit model with the consideration set composed of itineraries; ii) a PS logit model with the consideration set composed of common lines as part of the same alternative; and iii) a latent class model with consideration sets for two types of passengers: those who use itineraries and those who use common lines as part of the same alternative. We then use the results of these models to answer three research questions: Are public transport passengers using different route choice strategies? Is it possible to capture heterogeneity in route choice strategy by using latent class models estimated from smart card data? Can different behavioral parameters be captured?

Regarding the availability of data to observe route choice in public transport, in recent years some studies have generated data by observing the transactions registered in automatic fare collection systems through SC. If the card ID is available, panel data with multiple choice situations can be built from SC transactions. Unlike traditional data obtained from surveys, this type of passive data has negligible costs and allows the collection of large volumes of personal travel data over long periods of time (Bagchi & White, 2005). These advantages make SC data attractive for route choice studies. Previous studies that use SC data for route choice modeling include the estimation of stop-to-stop route choice models in different types of networks: i) a railway or metro network (Kusakabe et al., 2010; Van Der Hurk et al., 2015; Zhao et al., 2017); ii) a bus network, using a reduced number of origin–destination pairs (Nassir et al., 2018; Schmöcker et al., 2013) and iii) a large scale multimodal transit network (Jánošíková et al., 2014; Tan et al., 2015; Yap et al., 2020; Kim et al., 2020). The current study provides a new case study, using SC data to observe the route choice of passengers in a large-scale multimodal network with more than 20,000 origin–destination pairs.

From the smart card data, it is possible to obtain a large amount of information about a trip, such as the travel time in-vehicle, travel time out-of-vehicle, and the number of transfers. Additionally, combining AVL data, SC data, and GTFS data, it is possible to estimate the walking time in transfers and the waiting times at the beginning of the trips and the transfer points. Most studies that use SC data in route choice modeling assume that users maximize a utility function consisting of, mainly, the in-vehicle travel time, out-of-vehicle travel time, number of transfers, and an error term (Schmöcker et al., 2013; Jánošíková et al., 2014; Tan et al., 2015; Yap et al., 2020; Kim et al., 2020). The error component plays an important role in the route choice of passengers since it allows to explain the

variation of the chosen route through the passengers, even when they follow the same strategy. This is because the error term captures the non-added attributes such as the level of crowding, the type of buses, the preference for traveling with other users, availability of seats, individual characteristic, among others.

The remainder of this paper is organized as follows. The next section presents a literature review on route choice strategy. An analysis of the observed passengers' travel behavior for the current case study follows. Then, the paper discusses the proposed methods used. The model estimation results are then introduced. The last section draws conclusions and discusses policy and research implications.

2. Problem description

Every day, commuters and other travelers face the problem of selecting a route to arrive at a destination from a certain origin. To represent this choice, deterministic and probabilistic modeling approaches can be used. The deterministic approach is based on the minimum cost function, while the probabilistic approach incorporates unknown effects by adding an error term to the cost or utility function.

In the group of deterministic route choice models, the most common is the shortest-path heuristic (Gallo & Pallottino, 1988), where individuals are assumed to choose the itinerary with the lowest cost from the origin to destination. In the more complex hyperpath approach (Nguyen & Pallottino, 1988), passengers are assumed to consider a set of alternative routes that minimizes the total expected travel time and use the first line (from that set) that arrives at the bus stop.

Probabilistic modeling approaches are based on the Random Utility Maximization (RUM) framework, which has been widely used for representing route and mode choice behavior in the transport area (Mcfadden, 2000). The utility of each alternative is defined as depending on the valuation of a set of attributes through a set of coefficients that adjust to the perception of users, and a random component that represents the heterogeneity of preferences across individuals and additional factors unknown to the analyst.

Prato (2009) states that the major challenges in route choice modeling are the generation of a choice set of alternative routes and the estimation of discrete choice models. The first challenge is to identify the actual choice set considered by passengers, which is unknown to the researcher. The number of alternative routes is usually large and increases in a combinatorial dimension in large-scale networks. Therefore, identifying realistic choice sets is not a trivial task. The second challenge is to represent the actual behavior of travelers while appropriately considering the correlation between alternatives that share links and stops, and to estimate the coefficients of the different attributes using real data. As mentioned in the introduction, most literature on public transport passengers' route choice behavior has used the MNL model (Grison et al., 2017; Guo, 2011; Jánošíková et al., 2014; Nassir et al., 2018; Raveau & Muñoz, 2014; Raveau et al., 2014; Raveau et al., 2011; Vrtic & Axhausen, 2002; Kim et al., 2020). Only a handful of studies have actually considered similarity across routes via a Path Size logit (Anderson et al., 2017; Bovy & Hoogendoorn-Lanser, 2005; De Grange et al., 2012; Hoogendoorn-Lanser et al., 2005; Nielsen et al., 2021; Tan et al., 2015; Yap et al., 2020), possibly because of the complexity as well as the unimodality that reduces significantly the similarity across alternatives (especially for studies focusing only on rail or metro modes of public transport).

2.1. Basic definitions

In this section, we provide a formal definition for eight terms that we frequently. It is important to note that a node can be a bus stop or a metro/train station.

Trajectory: a sequence of public transport vehicles' positions over space.

Transit line/ line: is a group of public transport vehicles that run between an initial node and final node on the transit network. All vehicles in the same line travel on the network going always through the same sequence of network links and nodes. Each transit line provides a public transportation service defined by a sequence of stops, trajectory, and vehicle frequency.

Route: is a path that a passenger can follow on the transit network to travel between an origin and a destination zone. It can be identified by a sequence of nodes, the first one located in the origin zone of the trip, the final one being in the destination zone of the trip and all intermediate nodes represent the transfer points.

Route section: is a portion of a route between two consecutive initials, final and transfer nodes. Each route section is associated with a transit line or a group of transit lines.

Itinerary/ disaggregated alternative: is a route defined by a specific sequence of initial-transfer-final nodes with one specific line between each pair of consecutive nodes.

Aggregated alternative: is a route defined by a specific sequence of initial-transfer-final nodes with the group of common lines, between each pair of consecutive nodes, aggregated into a single alternative.

Common lines: the set of transit lines that altogether minimize the total expected travel time between two nodes in the network, i.e., if the passenger takes the first bus from this set of lines, the summation of the expected waiting time and the in-vehicle travel time will be minimal.

Transfer: the movement of a passenger from one public transport vehicle to another.

2.2. Route choice strategy

The literature suggests that passengers can adopt different strategies to choose a route to reach a destination (Raveau & Muñoz, 2014; Spiess & Florian, 1989). As we will see, the strategies can be used in both probabilistic and deterministic approaches.

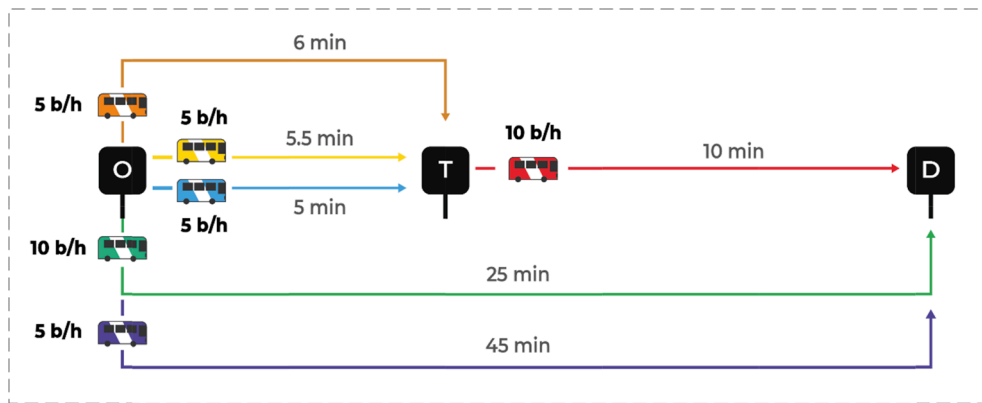


Fig. 1. Example of the route choice problem.

Table 1

Route alternatives using a disaggregated strategy.

Disaggregated alternative	In-bus travel time [min]	Expected waiting time [min]	Total travel time [min]
O - - T - - D	16	18	34
O - - T - - D	15.5	18	33.5
O - - T - - D	15	18	33
O - - D	25	6	31
O - - D	45	12	57

Table 2

Route alternatives using an aggregated strategy.

Aggregated alternative	In-bus travel time [min]	Expected waiting time [min]	Total travel time [min]
O - / / - T - - D	15.5	10	25.5
O - - D	25	6	31
O - - D	45	12	57

We propose to classify the strategies in two types: (i) a **disaggregated strategy**, where passengers choose a specific sequence of initial-transfer-final stops while considering specific lines between them (itineraries); and (ii) an **aggregated strategy**, where passengers choose a specific sequence of initial-transfer-final stops while aggregating the common lines (i.e., a set of lines that minimizes the total expected travel time) of each route section into a single alternative.

A graphical representation of a simple network that will help us to illustrate the two strategies is provided in Fig. 1, where six transit lines operate to serve one origin–destination (OD) pair with one transfer stop (T). The transit lines are shown in different colors with their respective frequencies [buses/hour] and travel time [min].

A disaggregated strategy in Fig. 1 would be: “Take the blue line from stop O to stop T, transfer to the red line and alight at stop D”. As can be seen, there are five itineraries or disaggregated alternatives for reaching destination D from origin O, which are described in Table 1. For each bus line, if the headway follows an exponential distribution, the waiting time is calculated as 60 min divided by the corresponding frequency. In-bus travel times are shown in Fig. 1. The itinerary O-green line-D minimizes the total travel time; therefore, in a deterministic approach, passengers will choose this alternative. In contrast, in a stochastic approach each itinerary has a non-zero probability of being chosen because of non-observed considerations represented by the error term in the utility function.

An aggregated strategy for the problem in Fig. 1 would be: “Take the next bus from either the orange, yellow, or blue lines from stop O to stop T, transfer to the red line and alight at stop D”. As can be seen in Table 2, there are three aggregated alternatives in this network. The first route is O-orange/yellow/blue line-T-red line-D, where in the first stage there are three common lines, and passengers take the first line that arrives at stop O. The waiting time of the first stage is 60 min divided by the summation of the frequencies of the common lines (15b/h); in the second stage, it is 60 min divided by the frequency of the red line (10b/h), resulting in a total waiting time of 10 min. The in-bus travel time in the first stage is calculated as a weighted average of the line frequencies (5.5 min); in the second stage, it is 10 min, as shown in Fig. 1. For the second and third alternatives, the waiting time and in-bus travel time are calculated in the same way as the disaggregated alternatives. It can be noticed that the alternatives that connect stops O and D, are not aggregated into a single alternative, because the green line and the purple are not common lines for this OD pair. The lines that do not belong to the common lines set are part of the consideration set, but they are aggregated into a different alternative to the common lines alternative.

As the first alternative has the lowest total travel time, in a deterministic approach all passengers will choose that option, and a proportion of 5/15 of passengers will be assigned to the orange line, 5/15 of passengers will be assigned to the yellow line, and 5/15 of passengers will be assigned to the blue line. In contrast, in a stochastic approach, each route has a non-zero probability of being chosen because of non-observed considerations represented by the error term in the utility function. Therefore, if the probability of choosing route 1 is p_1 , the probability of taking each line (orange, yellow, or blue) is $p_1 \cdot \frac{5}{15}$.

In general, discrete route choice models in the literature have used a probabilistic disaggregated strategy approach. On the other hand, transit assignment models have used a deterministic aggregated strategy approach. To the best of our knowledge, only [Raveau & Muñoz \(2014\)](#) and [Schmöcker et al. \(2013\)](#) have considered aggregated alternatives within a probabilistic route choice model.

The hyperpath strategy can be seen as an extension of the aggregated strategy. [Raveau & Muñoz \(2014\)](#) found that only 4% of passengers followed the hyperpath strategy, evidencing a low usage frequency of this strategy. Consequently, the hyperpath strategy is not included in this work.

3. Data description and analysis

The analysis for this study was carried out using the Santiago, Chile multimodal public transport network, known as Transantiago. Santiago is the capital of Chile and has a population of over 7 million inhabitants. The public transport system serves roughly 50% of motorized trips and it is operated by headway scheduling; therefore, lines do not have fixed-time schedules. The fare system is fully integrated, with an almost flat fare between urban buses, Metro, and some rail services, allowing up to three trip legs within a two-hour time window. In a typical week, 3 million passengers use the system to make 25.5 million trips. The network includes 7 metro lines, more than 300 bi-directional transit lines, and one rail service. In this work, we use observations from frequent passengers that travel during morning peak periods (6:30–8:30 AM) on weekdays. Specifically, we select users that travel 15 days or more on the public transport system during May 2018, and that stay in the destination locations for at least two hours. In this way, we try to capture trips to regular activities such as work or study. The morning peak period is the most congested, more than 700 thousand trips per day can be observed, and hence the most interesting travel period from a behavioral and planning perspective.

Very detailed demand information was obtained from the automatic fare collection (AFC) system in Transantiago ([Gschwender et al., 2016](#)). The smart card *bip!* is the only accepted payment method. Passengers must validate when boarding a bus or entering a Metro station. No alighting validation is required for bus or Metro trips. Around 27% of passengers evade fare payment on buses. Bus stops with particularly high demand have an off-vehicle payment system called *zona paga* (payment zone), where passengers validate when they enter the bus stop area and then board any bus without further validation. The data is already processed to estimate the boarding and alighting position for all validations and the trips (stages) associated with an origin–destination journey using the [Munizaga & Palma \(2012\)](#) methodology.

Additionally, in the Santiago public transport system, all buses are equipped with GPS devices that record the time and position every 30 s and we have used those data (AVL data) to obtain the observed frequency on transit lines.

3.1. Common lines analysis

To determine if passengers are using common lines, we identified, for each Origin-Destination (OD) pair, all route sections of the routes registered in the trips database. For those route sections, we identified the list of transit lines passing through and then applied the greedy heuristic ([Chriqui & Robillard, 1975](#)) to identify the set of common lines for each route section.

The optimization problem to find common lines is shown in Eq. (1), where the first term of this equation corresponds to the waiting time and the second term corresponds to the average of the in-vehicle travel times. [Spiess & Florian \(1989\)](#) explain that the waiting time parameter (α) can take a value of 1, when assuming an exponential distribution of interarrival times of vehicles, or a value of 0.5 when assuming constant headways of vehicles. Please note that different values of alpha parameter would result in different common line sets and consequently different aggregated strategies. In this study, we use the alpha parameter equal to 1 because previous studies have found irregular headways in the vehicles of the public transport system of Santiago ([Arriagada et al., 2019](#); [Godachevich & Tirachini, 2020](#)), and because the exponential distribution has been found to fit real waiting time data well ([Guo et al., 2011](#)). Specifically, in Eq. (1), it is assumed that buses arrive following a Poisson distribution with arrival rates being the sum of the observed frequency of lines, f_l being the observed frequency of line l , t_l being the in-vehicle travel time of line l , L being the number of lines serving the evaluated route section, and x_l taking value 1 when line l belongs to the common lines set and taking value 0 in other cases. Therefore, the optimal strategy for a user would be to take the first bus from this set of lines that arrives at the stop. In summary, the process to identify the common line set considers an evaluation of waiting time and in-vehicle travel time to classify a line as attractive (belongs to the common line set) or non-attractive. Therefore, common lines may or may not overlap in their trajectory, there are cases without overlap, with partial overlap, or with 100% overlap between lines that belong to the common line set.

$$\min \frac{\alpha}{\sum_{l \in L} f_l} + \sum_{l \in L} \frac{f_l t_l x_l}{\sum_{j \in L} f_j x_j} \quad (1)$$

Following this approach, we found 45,089 route sections with at least one trip. Among these, we analyzed 30,385 pairs, avoiding those with extra-vehicle payment systems (*zona paga*) and Metro stations, where the specific trajectory used is unknown because of the way in which it is inferred from the SC data ([Munizaga & Palma 2012](#)). For 95.8% of the route sections, all transit lines passing through were common lines and concentrated 468,275 of the observed trips. In the remaining 4.2% of route sections, which included 44,612

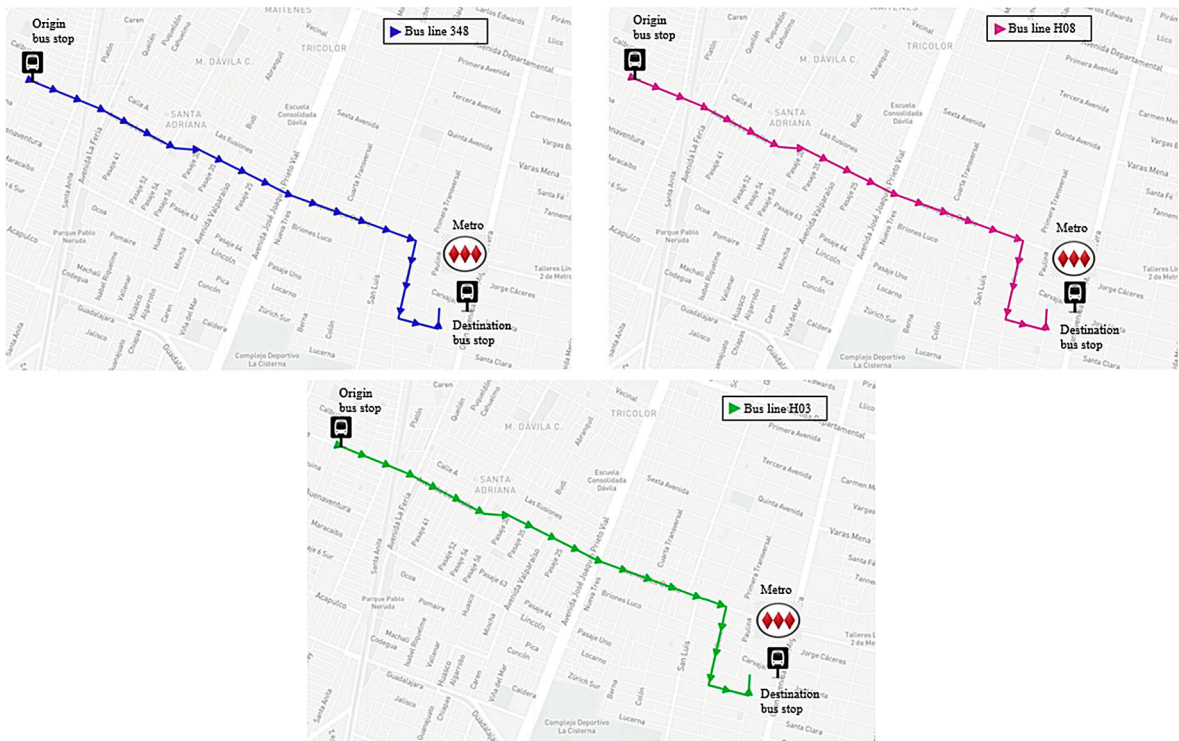


Fig. 2. Trajectory of the three common lines (348, H08 and H03).

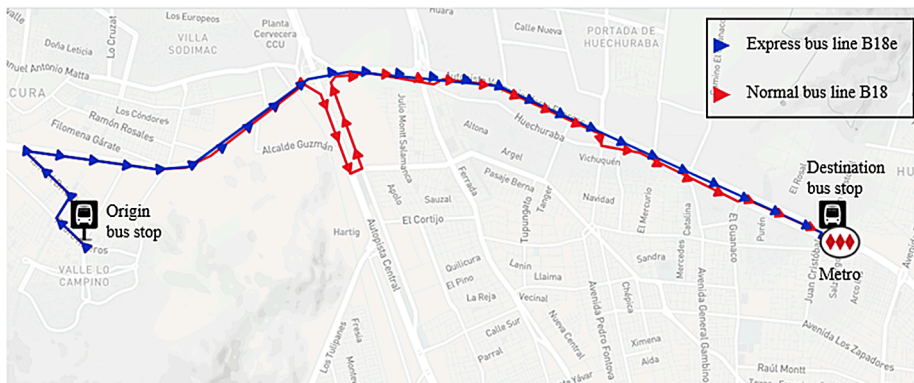


Fig. 3. Trajectory of the two non-common lines (B18e, express bus line and B18, normal bus line).

trips, some lines did not belong to the common lines set; in those cases, 79.9% of the trips were made using a line from the common lines set and 20.1% took buses from lines that did not belong to the common lines set.

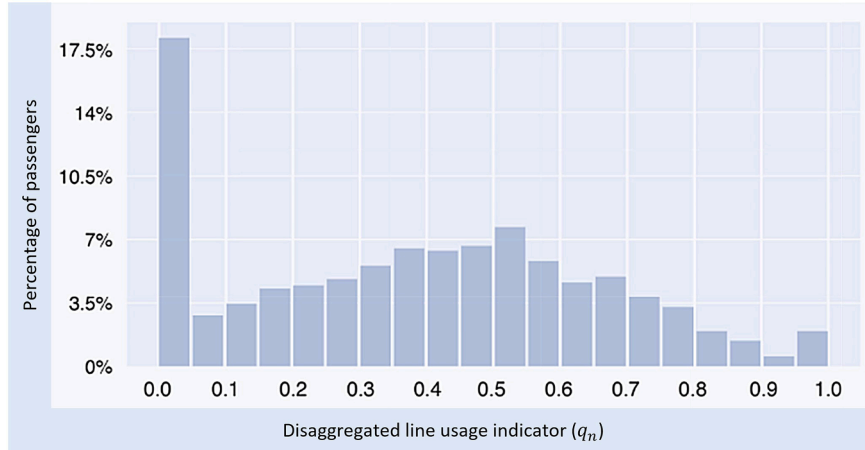
There are many reasons why a user would not follow the common lines strategy and end up considering instead a seemingly “slower” line for a trip. This could result from the omission of variables that are relevant for the user, such as (i) level of crowding, (ii) seat availability, (iii) the type of bus, as some lines may have newer buses or buses with air-conditioned or more comfortable seats, (iv) the bus operator, as some of them are more reliable. It could also result from behavioral conditions, such as (v) that the waiting time disutility can be higher than travel time disutility, (vi) or group behavior when passengers consider the utility of another passenger as part of the decision process. Any of these, or a combination of them, can make users prefer a slower line. We use the traditional version of the common lines approach that considers the summation of waiting and travel time, not assigning different weights to them. Possible extensions are left for further research, including the way in which the common lines set is built, that could consider a different valuation of travel time and waiting time, or incorporate other attributes.

As an illustrative example, we analyzed two route sections in the system in greater detail: one with common lines only (Fig. 2), and another with two non-common lines (Fig. 3). We compared the observed trip distributions with the expected trip distributions. For the expected trip distribution we assume that (i) passengers would board the first line belonging to the common lines set that arrived at the

Table 3

Statistics of lines in specific cases of common lines and non-common lines.

Case	Line	Travel time	Expected waiting time	Observed trips	Expected trips
Common lines	348	13.3 min	12.8 min	40 (26.0%)	49 (31.8%)
	H08	13 min	12.7 min	47 (30.5%)	49.4 (32.1%)
	H03	14.2 min	11.3 min	67 (43.5%)	55.6 (36.1%)
Non-common lines	B18	35.8 min	12.2 min	9 (4.8%)	0 (0%)
	B18e	17.5 min	7.8 min	179 (95.2%)	188 (100%)

**Fig. 4.** Distribution of disaggregated line usage indicator.

bus stop, and (ii) they would not board the lines that did not belong to the common lines set. The expected number of trips in a pair of stops or in a route section r along transit line l is expressed in Eq. (2), where OT_r is the observed total number of trips in route section r and P_{lr} is the probability of boarding line l to travel along the route section r . P_{lr} is shown in Eq. (3), where L_r is the number of common lines serving route section r .

$$ET_{lr} = P_{lr} OT_r \quad (2)$$

$$P_{lr} = \frac{f_l}{\sum_{i \in L_r} f_i} \quad (3)$$

In the first route section, shown in Fig. 2, the origin bus stop is connected to a Metro station by three common lines (348, H08, and H03) with in-vehicle travel times between 11.7 and 12.8 min and waiting times between 13 and 14.2 min. Table 3 shows the observed distribution and the expected distribution of trips among the three common lines. The expected distribution according to the common lines theory is not significantly different from the observed distribution (Pearson's chi-squared test), so in this example, we can conclude that passengers took lines according to the common lines principle. An implication of this finding is the possibility that passengers consider common lines as part of the same alternative since they take the first line that arrives at the stop.

In the case of the route section shown in Fig. 3 that has two lines, the origin bus stop is the first stop for two transit lines and is located in a suburban area. The only options to reach a Metro station are the local bus line (B18) or the express bus line (B18e). The optimal strategy is to wait for the express line (B18e), therefore line B18e belongs to the common lines set while B18 does not. According to the common lines model, all users should take the express line. However, Table 3 shows the observed distribution of trips, where 95.2% of trips are made with the express line and 4.8% are made with the local line. The 4.8% of trips made with the slower travel option were made by one passenger, who traveled in the analyzed OD pair 15 times, taking the local route 9 times and the express route 6 times. A possible explanation for this seemingly odd behavior might be any of the reasons explained above. This simple example suggests that lines that do not belong to the common lines set according to the model can still be attractive to some passengers.

In order to capture the behavior of passengers regarding the use of common lines within the database as a whole, we propose a disaggregated line usage indicator q_n . This indicator is based on the difference between the expected number of trips (proportional to the observed frequency of the common lines and equal to zero for transit lines not belonging to the common lines set) and the observed trips in a route section.

The disaggregated line usage indicator q_n is shown in Eq. (4), which has the purpose of measuring the degree on which the trips made by user n are concentrated in a single line or follow the common lines principle. The first summation adds over all the route sections visited by the user and the second sums over each common line belonging to the route section to compare the observed trips and expected trips in each common line. In Eq. (4), R_n is the number of the route section visited by passenger n , L_r is the number of common

lines serving route section r , and π_{lrn} is defined in Eq. (5). The elements in Eq. (5) are: the expected number of trips for user n on line l along the route section r (ET_{lrn}) and the observed number of trips for user n on line l along route section r (OT_{lrn}). The disaggregated line usage indicator q_n takes a value of 1 when all trips are concentrated in one of the common lines, or when they are made in some of the non-common lines, implying, to some degree, full adherence to the disaggregated strategy. In turn, q_n takes a value of 0 when the number of expected trips matches exactly the number of observed trips, implying perfect adherence to the common lines strategy. Values of q_n between 0 and 1 can be the result of random variations, measurement errors or that users combine aggregated and disaggregated strategies in their route choices. The number of observed trips is obtained from SC data, and the number of expected trips is calculated with Eq. (2).

$$q_n = \frac{\sum_{r=1}^{R_n} \sum_{l=1}^{L_r} \pi_{lrn}}{\sum_{r=1}^{R_n} L_r} \quad (4)$$

$$\pi_{lrn} = \frac{|ET_{lrn} - OT_{lrn}|}{\max(ET_{lrn}, OT_{lrn})} \quad (5)$$

Using real data, we calculated the disaggregated line usage indicator q_n for the users that traveled in route sections that do not begin in a *zona paga* or in a Metro station. We analyzed 28,896 cards (each corresponding to an individual user) and, as can be seen in Fig. 4, 20.9% of them (the sum of the first two bars) showed a value of q_n equal to or less than 0.1, which means that their observed behavior is consistent with an aggregated strategy for route choice, reflected in boarding the first bus of the common lines that arrive at the stop. Around 3% of them (the sum of the last two bars) showed a value of q_n equal or higher than 0.9, adhering with the disaggregated strategy, which means that they take one specific bus line, even when there are other common lines available. In summary, the analysis of q_n shows that some (20.9%) users use common lines, others use a subset of common lines, and others do not use common lines.

With this analysis, we have provided empirical evidence that passengers show heterogeneity in their route choice behavior. This finding has significant implications for route choice models, since these should incorporate heterogeneity across passengers in the use of alternative choice strategies.

4. Assessment of discrete choice modeling approaches

In this section, we study the research hypothesis by developing random utility maximization (RUM) models that incorporate disaggregated and aggregated route choice strategies.

4.1. Consideration set construction

The first challenge in the formulation and estimation of a public transport route discrete choice model from SC data corresponds to the construction of the consideration set. For the consideration sets, we first defined the origin and destination locations, which we considered as the areas within a 100-meter radius of the origin and destination stops of the trip. It is important to note that a sensitivity analysis was performed to define the radius of the origin and destination zones and we found that 100-meter radius zones generated the most consistent results in the models. In particular, larger radius zones generated unexpected results in the vehicle travel time coefficient. For each OD pair, we constructed the consideration set by combining all available and observed routes used by every passenger traveling within the defined origin and destination areas in the study period (6:30 to 8:30). It means that for each OD pair the same consideration set is used for all passengers.

The universal consideration set was built depending on the behavioral assumption of the respective model. As was explained before, we considered two possible approaches: (i) an aggregated strategy that implies aggregating the common lines set in a single alternative and taking the first bus that arrives at a stop and (ii) a disaggregated strategy that implies choosing only one line in each route section.

Fig. 1 illustrates an hypothetical OD pair, Table 1 shows the consideration set that would be built under a disaggregated strategy, and Table 2 shows the consideration set that would be built under an aggregated strategy. It is shown that if we construct the consideration set with aggregated alternatives, there are three possible routes. For route section O-T the orange line, yellow line and grey line belong to the common lines set, so they are part of the same alternative, and therefore there is only one alternative in the sequence O-T-D. For segment O-D there are two non-common lines, and therefore two alternatives. If we construct the consideration set with disaggregated alternatives, there are five possible routes in the consideration set.

4.2. Route choice modeling

As mentioned in the introduction, most route choice models assume that one strategy applies to all passengers within the network. However, this is not consistent with what we observe in the data, since the analysis using the disaggregated line usage indicator, in section 3, indicates passengers present different route choice strategies, some of them consider common lines and take the first incoming bus and others prefer to wait for a specific transit line. Therefore, our suggestion is to consider heterogeneity in the strategies used by passengers within route choice models. A latent class model (Walker, 2001) is appropriate for this route choice model because it incorporates different underlying behavioral rules. In our case, these behavioral rules dictate different ways of identifying the consideration set.

The latent class model has two components: a class membership model and a class choice model. In our context, there are two classes: passengers who consider common lines, and therefore use an aggregated strategy (class 1), and passengers who do not consider common lines, and therefore use a disaggregated strategy (class 2). The membership probability is modeled using a binary logit model, where the deterministic utility function v_n is interpreted as the propensity of passenger n to belong to the disaggregated strategy class. The propensity of the aggregated strategy class is fixed to zero for normalization purposes. The representation of the propensity function depends on a constant and on the dispersion γ_n of transit lines used by the passenger, as shown in Eq. (6). The dispersion can be measured with indicators based on the chi-square test, the Kolmogorov-Smirnov test, the Gini coefficient, and the statistical inference algorithm proposed by Nassir et al. (2017). In this study, we use the Gini coefficient (Glasser, 1962), which varies between 0 and 1. The value 0 means that the passenger made the same number of trips on each transit line that belongs to the common lines set, and the value 1 means that the passenger made all his/her trips on one line. Passengers with a Gini coefficient equal to zero have a higher probability of using common lines as a route choice strategy. The Gini coefficient is shown in Eq. (7), where R_n is the number of route sections visited by passenger n , L_r is the number of common lines serving route section r , OT_{lm} is the observed number of trips for user n on line l along route section r , and μ_m is defined in Eq. (8). The probability $\Lambda_n(\overline{CL})$ of belonging to class 2 is shown in Eq. (9), and the probability that passenger n selects alternative i is presented in Eq. (12), which is expressed using the total probability theorem.

$$v_n = \beta_{\overline{CL}} + \beta_\gamma \gamma_n \quad (6)$$

$$\gamma_n = \frac{\sum_{r=1}^{R_n} \frac{1}{2 \times \mu_m \times L_r \times (L_r - 1)} \sum_{l=1}^{L_r} \sum_{j=1}^{L_r} |OT_{lm} - OT_{jm}|}{R_n} \quad (7)$$

$$\mu_m = \frac{1}{L_r} \sum_{l=1}^{L_r} OT_{lm} \quad (8)$$

$$\Lambda_n(\overline{CL}) = \frac{\exp(v_n)}{\exp(v_n) + 1} \quad (9)$$

Since we do not know if a passenger uses common lines due to the latent nature of the behavior (i.e., it is not observed), we try to approximate this behavior through the indicator γ_n , which cannot be put directly into the systematic utility function because it would result in endogeneity due to the correlation between γ_n and the error term (Guevara, 2015). This problem can be solved with the latent class approach (Guevara, 2015). However, it requires formulating a structural equation and resolving the problem through maximum simulated likelihood, which is a memory-intensive process and requires more time than any other part of the algorithm that is already computationally very expensive. To avoid this problem, we address the endogeneity problem using the Multiple Indicator Solution (MIS) method that was adapted to discrete choice models by Guevara & Polanco (2016).

The MIS method is proposed as an accessible way to address endogeneity when it is difficult to gather traditional instruments, but two (or more) indicators of the omitted variable causing the endogeneity are available. In that case, it can be shown (see Guevara and Polanco, 2016) that the problem will be solved if the researcher adds one of the indicators as an auxiliary variable of the model and uses the other indicator(s) as instrument(s) for applying the Control Function method. The intuition behind the MIS method is that, by including the first indicator as an auxiliary variable, one eliminates the endogeneity produced by the omitted attribute, but, in turn, causes a different type of endogeneity resulting from the use of an improper proxy. Then, in this modified model, the only remaining source of endogeneity resides in a measurement error for the included indicator, which can be solved using the other indicator(s) as instrument(s) in a Control Function application.

The MIS method is applied using the disaggregated line usage indicator q_n (Eq. (4)) as the instrument, and the Gini coefficient as the indicator, in two stages. The first comprises an OLS regression of the Gini coefficient on its instrument, the disaggregated line usage indicator q_n (Eq. (10)). The second stage consists of adding $\hat{\delta}_n$, the residual of Eq. (10), to the propensity equation to correct for endogeneity, as shown in Eq. (11). Finally, the choice model is estimated using the choice probability shown in Eq. (12) but using the systematic propensity v_n shown in Eq. (11).

$$\gamma_n = \beta_{\text{intercept}} + \beta_q q_n - > \hat{\delta}_n \quad (10)$$

$$v_n = \beta_{\overline{CL}} + \beta_\gamma \gamma_n + \beta_\delta \hat{\delta}_n \quad (11)$$

$$P_n(i) = P(i|CL)(1 - \Lambda_n(\overline{CL})) + P(i|\overline{CL})\Lambda_n(\overline{CL}) \quad (12)$$

Given the consideration set C_n for passenger n , we adopt a RUM framework, where we associate a utility for each route alternative and choose the route with the highest utility value. The utility has two components: the random component, which is assumed to be Gumbel distributed; and the deterministic component, which is specified in Eq. (13).

$$V_i = \sum_m \beta_{tm} tt_{mi} + \sum_c \beta_{tc} t_{ci} + \sum_s \beta_{tr_s} tr_{si} + \beta_{PSC} PSC_i + \beta_{CR} CR_i BUS \quad (13)$$

In this equation, i represents the alternative route, m is the public transport mode ($m \in \{bus, metro\}$), tt_{mi} is on board travel time, which is included in the sum over all modes of the alternative i . t_{ci} represents other time components ($c \in \{\text{initialwaitingtime(IWT)}, \text{transferwaitingtime(TWT)}, \text{transferwalkingtime(TWaT)}\}$), it is included in the sum over all components

times of the alternative i . tr_{si} represents the penalties for different transfer types ($s \in \{\text{bus} - \text{to} - \text{bus}(\text{BB}), \text{Metro} - \text{to} - \text{bus}(\text{MB}), \text{bus} - \text{to} - \text{Metro}(\text{BM})\}$), it is included in the sum over all transfer types of the alternative i . BUS is a binary variable that takes a value of 1 when the alternative contains the bus mode, and CR_i is the level of crowding at line-stops that are used within the alternative. PSC_i is the path size correction term to capture the correlation due to overlapping between alternative routes.

The level of crowding of a line in a route section is calculated as the number of passengers inside of buses divided by the capacity of the buses. The level of crowding in an alternative route is the average of the crowding of each route section that belongs to that alternative.

Path size correction introduces a negative factor (PSC_i) that decreases the deterministic utility of alternative routes that have correlation with other routes. We have used the expression according to Bovy et al. (2008) in Eq. (14), where L_r is the length of the route section r , L_i is the length of route i , ζ_i is the set of route sections belonging to route i , and δ_{rk} is the route section-route incidence number, which takes a value of 1 if route k uses route section r and a value of 0 otherwise.

$$PSC_i = \sum_{r \in \zeta_i} \left(\frac{L_r}{L_i} \ln \frac{1}{\sum_{k \in C_p} \delta_{rk}} \right) \quad (14)$$

The waiting time included in t_{ci} in Eq. (13) is obtained assuming an exponential distribution as one divided by the observed frequency of the line (or sum of frequencies, in the case of common lines). The exponential distribution (or the gamma, which is the sum of exponentials) has been widely used for modeling waiting time in transportation systems (see, for example, Nguyen & Pallottino, 1988; Raveau & Muñoz, 2014; Schmöcker et al., 2013).

In aggregated alternatives, the time components and level of crowding per route section are a weighted average (based on frequency) of the values corresponding to the transit lines that conform each route section.

Because of the Gumbel distribution of the error term, the probability of passenger n choosing itinerary i given consideration set C is expressed as a PS logit model (Bovy et al., 2008). Eq. (15) presents the probability of choosing itinerary i in the case of a passenger who does not use common lines and Eq. (16) presents the probability of choosing itinerary i in the case of a passenger who uses common lines, where $V_{j(i)}$ is the systematic utility of the aggregate alternative route j which contains itinerary i , ζ_i is the set of route sections that belong to alternative route i , L_r is the set of common lines that belong to route section r , and $f_{l(i),r}$ is the observed frequency of transit line l which passes through route section r and is part of itinerary i .

In the last case, it is necessary to multiply the probability of choosing the aggregated alternative with the probability that the line, in each route section, arrives first at the stop. As an example, if we want to know the probability of choosing itinerary alternative *O-orange line-T-red line-D* in Fig. 1, and the passenger uses common lines, we should multiply the probability of choosing aggregated alternative *O-orange/yellow/blue line-T-red line-D* by the probability that the orange line arrives first at stop O, which is proportional to its observed frequency.

$$P(i|\overline{CL}) = \frac{\exp(V_i)}{\sum_{k \in C} \exp(V_k)} \quad (15)$$

$$P(i|CL) = \frac{\exp(V_{j(i)})}{\sum_{k \in C} \exp(V_k)} \prod_{r \in \zeta_{j(i)}} \frac{f_{l(i),r}}{\sum_{k \in L_r} f_{k,r}} \quad (16)$$

The closed-form logit formula of the PS logit allows for a simple estimation of the fixed coefficients by maximizing the likelihood function.

4.3. Consideration sets results

The alternative routes were generated using historical choices for each OD pair. As explained in previous sections, we worked with two types of alternatives: disaggregated alternatives were used to construct consideration sets without common lines, and aggregated alternatives were used to construct consideration sets with common lines. We worked with OD pairs that featured between 2 and 10 disaggregate alternatives; OD pairs with fewer or more disaggregate alternatives were discarded. Using this filter, we obtained 20,871 OD pairs with an average of 3.45 available alternatives within the consideration set with disaggregated options, and 1.38 available alternatives with aggregated options.

The overlap between alternative routes was evaluated at the stop level, which means that the alternatives that used the same route sections and transport mode were considered as correlated elements. It is important to note that for both, disaggregated and aggregated alternatives, the Path Size correction term was obtained with the same procedure, described in Eq. (14). In order to compare the correlation of each link with other alternative routes, we derived transit line trajectories from GTFS data (for Metro, we did not know the trajectory used by passengers, so we assumed that passengers took the route with the lowest travel time). The Path Size correction term (PSC_i) takes a value of 0 when there is no overlap between the evaluated alternative and others, and the value decreases as the level of route correlation increases; that is, the smaller the PSC_i value, the higher the correlation of route i . Consideration sets without common lines have a mean PSC_i of -0.85 and choice sets with common lines have a mean PSC_i of -0.16 . The lower value (more negative) in PSC_i for disaggregated alternatives is expected because the conformation of alternatives with common lines captures a part of correlation which is considered by PSC_i in disaggregated alternatives.

Table 4

Statistics of attributes used in PS logit and latent class models.

Attribute	Disaggregated alternatives			Aggregated alternatives		
	Min	Mean	Max	Min	Mean	Max
Travel time in bus (t_{tbus})	0	16.55	89.53	0	16.55	89.53
Travel time in metro (t_{tmetro})	0	17.68	72.73	0	17.68	72.73
Initial waiting time (t_{tWT})	1.7	8.33	17.47	0.81	4.18	17.46
Transfer waiting time (t_{tWT})	0	2.90	17.5	0	2.38	17.40
Transfer walking time (t_{tWT})	0	0.78	9.79	0	0.82	9.79
Transfer bus-bus (t_{tBB})	0	0.08	2	0	0.08	2
Transfer Metro-bus (t_{tMB})	0	0.05	1	0	0.05	1
Transfer bus-Metro (t_{tBM})	0	0.62	1	0	0.62	1
Total transfer	0	0.75	2	0	0.75	2
Bus crowding level (CR)	0	0.21	1.2	0	0.21	0.95

Table 5

PS logit estimates (t tests) and rates of substitution in case of disaggregate alternatives and aggregate alternatives. All columns show t-values between parentheses. t-tests are against zero for the model estimates and against one for the rates of substitution. For the rate of substitution, we followed the procedure proposed by [Daly et al. \(2012\)](#).

Parameters	Model estimates		Rates of substitution	
	Disaggregated alternatives	Aggregated alternatives	Disaggregated alternatives	Aggregated alternatives
Travel time in bus (β_{tbus})	-0.063 (-43.8)	-0.062 (-37.2)	1	1
Travel time in Metro (β_{tmetro})	-0.067 (-34.5)	-0.070 (-32.6)	1.06 (3.3)	1.13 (6.4)
Initial waiting time (β_{tWT})	-0.098 (-82.0)	-0.139 (-57.8)	1.56 (13.9)	2.24 (16.4)
Transfer waiting time (β_{tWT})	-0.082 (-24.2)	-0.080 (-11.3)	1.30 (5.0)	1.29 (2.4)
Transfer walking time (β_{tWT})	-0.309 (-44.6)	-0.091 (-9.3)	4.91 (27.0)	1.47 (2.9)
Transfer bus-bus (β_{tBB})	-1.017 (-19.3)	-0.727 (-12.1)	16.14 (16.2)	11.73 (10.3)
Transfer bus-Metro (β_{tBM})	-0.732 (-18.6)	-0.673 (-15.0)	11.62 (16.6)	10.84 (13.4)
Transfer Metro-bus (β_{tMB})	-0.948 (-6.7)	-1.000 (-6.8)	15.05 (6.2)	16.13 (6.3)
Bus crowding level (β_{CR})	-0.807 (-17.6)	-0.711 (-7.7)	12.81 (15.0)	11.47 (7.1)
PSC (β_{PSC})	-0.132 (-4.6)	0.071 (1.3)		
N° observations	150,418	150,418		
Log-likelihood	-157050.3	-35978		
Adjusted rho-square	0.044	0.083		

4.4. Model estimates

PS logit models and latent class models were estimated with a sample of 150,418 observations, which corresponds to trips made during 15 workdays during the morning peak period (between 6:30 and 8:30 AM). The PS logit was estimated in two cases, with aggregated and disaggregated alternatives routes. The latent class models were estimated with the MIS approach correcting for endogeneity and the two types of choice sets.

Please note that trip observations may have route sections served by only one transit line, and in those cases, the disaggregated line usage indicator takes the value zero, indicating that those passengers follow the common line principle. However, in these cases, the aggregated and disaggregated alternatives are the same. In this study, those observations, which correspond to 1.6% of passengers, are included for modeling purposes since even though they may have only one alternative in a route section, they have more than one alternative for the entire trip.

The specification of the deterministic utility function considers in-vehicle travel time for bus and Metro, waiting time at the beginning of the trip and during transfers, walking time during transfers, transfer penalties for bus to bus, bus to Metro, and Metro to bus, and the level of bus crowding. Metro to Metro transfers cannot be incorporated into the model because the information about the route that the passenger used inside the Metro network is not available. [Table 4](#) shows the minimum, mean, and maximum values for each attribute considered in the models. Aggregated alternatives present lower values for waiting time because common lines are considered as part of the same alternative.

[Table 5](#) shows the estimated parameters and rates of substitution for PS logit models using the consideration set with disaggregated and aggregated alternatives. The parameters are statistically significant and with the expected sign, except for the term PSC, which is not statistically significant in the aggregated alternatives model and is statistically significant but with the unexpected sign in the disaggregated alternatives model. Given that PSC belongs to the interval $(-\infty, 0]$ and implies a reduction in the systematic utility of correlated routes, we expect a positive sign in the coefficient. The rates of substitution suggest that transfers represent between 11 and 16 min of Travel Time in Bus (TTB), a 10% increase in crowding level represents between 1.2 and 1.3 min of TTB, travel time in Metro

Table 6

Multiple Indicators Solution latent class estimates (t tests) and rates of substitution in case of disaggregate alternatives and aggregate alternatives. All columns show t-values between parentheses. t-tests are against zero for the model estimates and against one for the rates of substitution. For the rate of substitution, we followed the procedure proposed by [Daly et al. \(2012\)](#).

Parameters	Model estimates		Rates of substitution	
	Disaggregated alternatives	Aggregated alternatives	Disaggregated alternatives	Aggregated alternatives
Travel time in bus ($\beta_{t_{bus}}$)	-0.085 (-42.7)	-0.05 (-19.1)	1	1
Travel time in Metro ($\beta_{t_{metro}}$)	-0.081 (-30.0)	-0.056 (-16.9)	0.95 (-2.1)	1.1 (3.4)
Initial waiting time ($\beta_{t_{wt}}$)	-0.041 (-24.6)	-0.099 (-29.9)	0.5 (-22.6)	2 (7.7)
Transfer waiting time ($\beta_{t_{trwt}}$)	-0.043 (-9.0)	-0.058 (-6.3)	0.5 (-8.6)	1.2 (0.8)
Transfer walking time ($\beta_{t_{trwt}}$)	-0.396 (-39.6)	-0.08 (-5.8)	4.7 (25.1)	1.6 (2.1)
Transfer bus-bus ($\beta_{t_{rbb}}$)	-0.974 (-12.0)	-0.944 (-11.5)	11.5 (10.3)	18.9 (9.1)
Transfer bus-Metro ($\beta_{t_{rbm}}$)	-0.938 (-14.0)	-0.722 (-12.1)	11 (12.6)	14.4 (10.4)
Transfer Metro-bus ($\beta_{t_{rmb}}$)	-1.349 (-6.8)	-0.85 (-3.4)	15.9 (6.3)	17 (3.2)
Bus crowding level (β_{CR})	-1.016 (-15.8)	-0.472 (-3.9)	12 (14.0)	9.4 (3.5)
PSC (β_{PSC})	-0.339 (-7.5)	0.189 (2.6)		
Class membership model				
Constant disaggregated strategy (β_{CL})	-19.25 (-7.4)			
Gini indicator (β_{γ})	50.22 (7.5)			
Residual (β_{δ})	-108.3 (-7.4)			
N° observations	150,418			
Log-likelihood	-155825.7			
Adjusted rho-square	0.031			

is more burdensome than travel time in bus, and initial waiting time, transfer waiting time and transfer walking time are more burdensome than travel time in bus.

We estimated a simple latent class model using Eq. (6) for the propensity of the disaggregated strategy, which adds the Gini coefficient to the utility specification. The results of this model showed unexpected results, such as a positive sign for the walking transfer time coefficient and transfer bus to bus coefficient in the class aggregate alternatives. These biased parameters can be explained because of the endogeneity, given that the Gini coefficient is correlated with the error term of the propensity function. For this reason, we applied the MIS method including the Gini coefficient in the membership model and disaggregated line usage indicator as the instrument, as described in Eq. (11). This model, shown in Table 6, is better than the endogenous model, since the coefficient of walking transfer time variable and transfer bus to bus variable have a negative sign. The fit of the MIS model is slightly better than that of the simple latent class model. This allows us to conclude that the combination of the Gini coefficient with the disaggregated line usage indicator provides more information than the latent class model without any indicator or with the Gini coefficient alone.

We tested the application of the MIS method including the disaggregated line usage indicator in the membership model and the Gini coefficient as the instrument; however, it obtained a non-statistically significant coefficient for the disaggregated line usage indicator, which can be explained because the Gini indicator presents poorer performance than the disaggregated line usage indicator ([Guevara et al., 2018](#)).

The rates of substitution in the MIS latent Class model (Table 6) confirm some findings within the PS logit models, but there are also important differences between them. In contrast to the PS logit models, we found the following results in the MIS latent Class model: both the initial waiting time and the transfer waiting time are perceived as less unpleasant than TTB for the disaggregated alternatives class, increasing the difference in perception of waiting time between both classes of passengers, given that individuals of the aggregated class find initial and transfer waiting times more unpleasant than TTB; bus-to-bus transfers are more unpleasant for passengers belonging to the aggregated alternative class; and Metro-to-bus transfers are more unpleasant for passengers belonging to the disaggregated alternative class.

The most unpleasant transfer for the disaggregated strategy class is Metro-to-bus, which represents 15.9 min of TTB, while for the aggregated strategy class the most unpleasant is bus-to-bus, representing 18.9 min of TTB. With respect to the bus crowding level variable, a 10% level of bus occupation represents 1.2 min of TTB for the class without common lines, and 0.9 min for the common lines class. On the other hand, waiting time at the beginning of the trip and during transfers is more burdensome for passengers who use aggregated alternatives, which is to be expected, as people consider common lines with the purpose of decreasing waiting time. Bus crowding levels and walking during transfers are more burdensome for passengers who use disaggregated alternatives; therefore, we can suggest that these passengers may not be using all common lines because they try to avoid walking and crowding situations. The membership class model parameters have the expected signs and are statistically significant, showing the heterogeneous route choice strategies of passengers.

We tested the stability of the parameters of the MIS latent class model with respect to the value of the waiting time parameter (α) to define the common lines, considering that different values would result in different aggregated strategies. We tested three values different from 1: $\alpha = 0.5, 0.75$ and 1.5 , to define the common lines that are considered for the aggregated alternatives' class. The results, shown in Table 7 in the Appendix, suggest that in all cases there was heterogeneity in the route choice strategies used by passengers, and the waiting time was relatively more onerous for passengers using aggregated alternatives. On the other hand, the models that

were built with α below 1 showed difficulties to adequately capture the effect of the level of crowding and, furthermore, the effect of the path size was only adequately captured for the aggregated class when α was equal to 1.

The adjusted rho-square of the models varies between 0.03 and 0.08. It should be reminded that rho-square in discrete choice models does not have a direct interpretation as explanatory power. It has been shown, e.g., by Bohara et al. (2007), it is possible to accept a model with a rho-square value of around 0.05. To confirm the validity of our models we performed a likelihood ratio test that rejected the null hypothesis that the coefficients are equal to 0, which shows that there is a gain in estimating the decisions of passengers using the proposed models.

5. Discussion

This section is devoted to the analysis of PS and MIS latent class models parameters with the purpose of understanding route choice behavior in a multimodal large-scale network.

5.1. Travel time

The negative coefficients of travel time variables show a disutility of travel time for passengers. In the models, we estimate travel time by mode to understand the sensitivity of travel time for individuals per mode. The results in both path size logit models indicate that passengers find travel time in Metro to be more burdensome. This is not surprising, as in Santiago, the Metro offers high frequency and reliable service (stable headways) but the crowding levels inside the cars are significant, reaching up to seven passengers per square meter during peak hours, making even short trips uncomfortable. The opposite was found by Eluru et al. (2012) in Montreal, and by Anderson et al. (2017) in Copenhagen, where the unobserved factors are different. In those cases, the Metro network, besides having a high frequency and stable operations, provides protection from extreme weather conditions, especially during winter. Therefore, users prefer to travel underground.

Furthermore, the latent class models indicate that there are differences between the two classes of passengers in terms of how they perceive travel time in bus and Metro. Passengers who choose disaggregated alternatives find travel time in a bus slightly more unpleasant than travel time in Metro, while passengers who choose aggregated alternatives find travel time in Metro slightly more unpleasant than travel time in a bus. A possible explanation for this difference in the effect of travel time is related to the physical discomfort in Metro. There is a need for future research to examine this difference in detail.

5.2. Waiting time

The models indicate that alternative routes with lower waiting times are preferred. In the models we estimated sensitivity to waiting time at the beginning of the trip as well as at any transfer stages. The results in all models (except the disaggregated alternatives in the MIS latent class model) indicate that passengers find waiting time at the beginning of the trip to be more onerous than waiting time during transfers. Few studies have reported this difference in the perceived waiting time, which can be explained because individuals feel particularly anxious to start the trip, while during transfers, this anxiety decreases since the trip has already begun.

Differences in waiting time perception are shown in the MIS latent class model, where passengers that choose aggregated alternatives find the waiting time more onerous than passengers who choose itineraries. This is expected, as passengers use common lines to decrease their waiting time. On the other hand, surprisingly, the waiting time is found to be less onerous than the in-vehicle time for disaggregated strategy class. It might be the case that users who care less about waiting time, are more willing to wait for a specific bus line of their preference (disaggregated strategy), which reflects in a smaller coefficient for waiting time than in-vehicle travel time. Likewise, it might be the case that users who use disaggregated alternatives are keener to use mobile transport applications that provide bus arrival time that allow them to further reduce their waiting time because they can arrive at the bus stop just in time to take the bus. As the model calculates the waiting time as a function of the observed frequency, yielding the same value for all users, the waiting time for the users who use mobile transport applications might be overestimated. If they are concentrated in the disaggregate alternatives class, this may bias our results for that class, reflecting in a smaller coefficient for waiting time than in-vehicle travel time.

5.3. Walking time

Routes with lower walking time during transfers are preferred. All models show that walking time is more onerous than travel time in bus, and the MIS latent class model indicates that individuals from different classes show a substantial difference in the effect of the walking time variable. For passengers who choose itineraries, walking time is more unpleasant than it is to passengers in the other class, which can explain why those passengers prefer a specific itinerary instead of all lines that belong to the common lines set.

5.4. Transfer

The alternatives used in this work involve transfers. When there is a potential waiting time and walking time for each transfer, we incorporated the influence of transfers on alternative routes in multiple ways: transfers per mode, transfer waiting time, and transfer walking time. As expected, alternatives with fewer transfers, lower waiting times, and lower walking time per transfer are preferred. As can be seen in the MIS latent class models, both types of passengers, from the disaggregated and aggregated strategy class, prefer the bus-metro transfer to other types of transfers (bus-bus, metro-bus). This can be explained because the metro is more regular than the

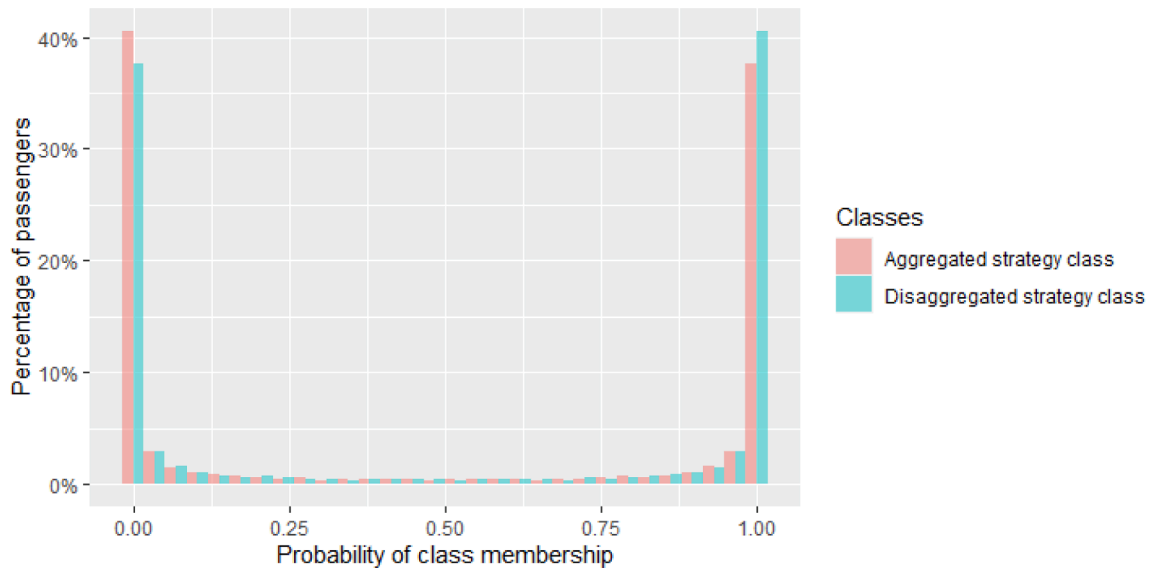


Fig. 5. Histogram for the estimated class membership probabilities for the MIS latent class model with Gini coefficient as the indicator and disaggregated line usage index as the instrument. $N = 18,466$.

buses, and then all passengers prefer to transfer to a more regular service. On the other hand, the most disliked type of transfer for passengers who choose disaggregated alternatives is Metro-to-bus; for passengers who use common lines, the most disliked type of transfer is bus-to-bus. The largest difference between classes occurs in the bus-to-bus transfer that jumps from 11.5 min in the disaggregated strategy class to 18.9 min for the aggregated strategy class. We speculate that this may occur because bus-to-bus transfers are prone to a larger uncertainty, which generates an extra difficulty to plan a trip when this type of transfer is considered as an alternative, especially in the case of passengers that plan a trip using aggregated strategy. This may result in a larger bus-to-bus transfer penalty for that class. This must be further analyzed with additional information.

5.5. Path size term

In the PS logit model with disaggregated alternatives, as with the disaggregated alternatives class in the MIS latent class model, the results showed that the path size correlation has a negative sign, which is not expected. Nevertheless, [Anderson et al. \(2017\)](#) obtained the same result for the correlation term, explaining this phenomenon as the fact that passengers prefer more opportunities to reach their destination from their point of origin. In this study, the correlation term was positive, as expected, in the aggregated alternatives.

It should be noted that the negative value for the path size is not uncommon in the literature about public transport traveler's route choice. In fact, although the path size is supposed to correct for overlapping routes by reducing the utilities of overlapping routes, negative estimates for path-size terms have been found ([Anderson et al., 2017](#); [De Grange et al., 2012](#); [Hoogendoorn-Lanser and Bovy, 2007](#)), most likely because of the additional utility of travelers having more opportunities to reach their destination from their origin, or because travelers might value the availability of a large number of en-route alternative options over the uniqueness of the route ([Anderson et al. 2017](#)). A very recent study ([Nielsen et al., 2021](#)) has even found non-significance of the path size estimate once transfer related variables were inserted explicitly in the utility function, suggesting that the inclusion of those variables captures the similarity explicitly.

5.6. Membership model of the latent class model

Using the Bayesian estimator of class membership ([Greene and Hensher, 2003](#)) in the MIS latent class model, we determined that the mean probability that the passengers belong to the disaggregated strategy class is 51.2% and that 48.8% is the mean probability that the passengers belong to the aggregated strategy class. [Fig. 5](#) shows a histogram for the predicted probabilities of belonging to both latent classes. It can be noted that the high degree of heterogeneity in the choice strategy occurs between individuals, not within the respective choices of each one. It can be seen that around 41% of individuals have a probability higher than 0.98 of belonging to disaggregated strategy class and other 38% have a probability higher than 0.98 of belonging to aggregated strategy class, which means that there is a similar proportion of passengers that wait for specific transit lines and that consider common lines as part of the same alternatives.

In order to understand if the distance of the trip generates an effect in the route choice strategy of passengers, we calculated the probability of class membership separately for those passengers who make short trips (10 km or less of Euclidean distance) and who make long trips (more than 10 km of Euclidean distance). The results of this analysis do not show evidence that trip distance influences

the choice of the route choice strategy. In this line, further work needs to be done to establish whether some characteristics of the OD pairs affect the route choice strategy of passengers. One example is the level of overlapping between alternative routes, which can allow distinguishing users who consider complex aggregated strategies (common lines with a small level of overlapping) versus users who consider less complex aggregated strategies (common lines with a high level of overlapping).

6. Conclusions

This study uses smart card data from Santiago, Chile to evaluate the route choice strategies of public transport system users. We identified two types of strategies in the existing literature: (i) a disaggregated strategy and (ii) an aggregated strategy.

To analyze the use of common lines (aggregated strategy), we used an indicator based on the difference between the number of expected and observed trips for each passenger within lines that connect the route sections used by the passenger to make trips. The results of this analysis showed evidence for heterogeneity within route choice strategies. Some passengers use common lines in each stage of their trip, while others use one specific line, even when there is more than one common line between their origin and destination. Another group is not well-defined, because it uses a subset of common lines, as defined by [Chriqui & Robillard \(1975\)](#).

Because our data showed that a proportion of passengers uses common lines in each stage of their trip, we estimated three RUM models: a PS logit model with disaggregated alternatives or itineraries, which assumes that all passengers use a disaggregated strategy; a PS logit model with aggregated alternatives, which assumes that all passengers use an aggregated strategy; and a MIS latent class model, which takes in consideration the endogeneity problem. The MIS latent class model contains two classes, disaggregated alternatives and aggregated alternatives, assuming that passengers may use a combination of aggregated and a disaggregated strategies. One of the more significant empirical findings to emerge from this study is that passengers present differences in their travel behavior when they choose a route to arrive to the destination: in general, some consider common lines, while others wait for a specific line.

It is interesting to note that both groups of passengers have differences in their perceptions of some route attributes. The MIS latent class model contributes to our understanding of those differences, which would not be possible to obtain with a simple multinomial logit model or with a PS logit model. With the MIS latent class model, we see that passengers using the aggregated strategy prefer to travel in bus rather than Metro, while passengers in using the disaggregated strategy have the opposite preference for one transport mode over the other. Waiting time is more burdensome for passengers who use common lines, lending support to why they consider more than one line as part of the same alternative. Walking time and bus crowding is more burdensome for passengers who use disaggregated alternatives, suggesting that they prefer specific lines in order to avoid walking and/or crowding.

With the estimation of the class membership parameters in the latent class model we found that the percentage of passengers that use itineraries as route alternatives is similar to the percentage of passengers who use common lines. We speculate that a high proportion of passengers use disaggregated strategy because a proportion of them do not know which transit lines belong to the common line set or they use real-time information to reduce their waiting time choosing the transit line that they prefer. This suggests that in order to reduce waiting times and improve passenger perception of public transport, it may be important that transport authorities make real-time information channels available, allowing users to know which common line alternatives allow them to reach their destination.

In this study, we use observations from frequent passengers that travel during morning peak periods for at least 15 times during a month, and that stay in the destination locations for at least two hours. In this way, we try to capture trips to regular activities such as work or study. A natural progression of this work is to analyze the route choice strategies using data from less frequent users, who probably do not know the transport network as well and, we speculate, may use more disaggregated strategies than aggregated strategies. In other words, future research is required to determine if the frequency of public transport use affects the route choice strategy of passengers. Other interesting extensions of this work would be to analyze other periods of the day, such as the afternoon peak period and off-peak hours, where one would expect that passengers have different preferences. We speculate that, for those periods, the disutility of the waiting time and travel time might be smaller since users do not have a tight schedule to reach their destination, as it happens in morning peak hours with trips to work or study.

Since the public transportation system in Santiago operates by headway scheduling, without fixed-time schedules, the analysis of route choice strategy undertaken here has extended our knowledge of route choice behavior of passengers that travel in frequency-based transit networks. In this context, it is important to address the common lines problem, because users face a choice between multiple transit lines from one stop to the next transfer station, and some of them prefer to board the first bus that arrives to minimize the total expected travel time. Several questions remain to be answered about the route choice strategy used by passengers that travel in other contexts. For example, passengers that travel in a schedule-based transit network, where users can optimize their trips by planning their arrival time to the bus stop. This can also be the case when users have real-time information about buses' arrival times. These contexts allow passengers to save waiting time (if the transit lines are punctual or the online information is accurate) while choosing their preferred lines. We might expect that in these cases, people may tend to use more the disaggregated strategy.

Another possible line of research is related to the improvement of the process of constructing common lines by considering different valuations for travel and waiting times, and/or by the incorporation of other attributes. An intuitive first step toward this effort could be to use for that the attributes' valuation that resulted from the choice modelling process. That approach would imply building an iterative process of uncertain convergence and huge computational costs but, more importantly, theoretical and empirical evidence suggests against it. [Swait and Ben-Akiva \(1987\)](#) show that it would be inappropriate to assume that attribute valuation obtained from the choice modeling stage is the same as those regarding the choice set formation, which is the role played by the common lines in our framework. Instead, if one would want to modify the valuation of the attributes of the common lines approach, the correct way to address it would have to be built within the framework of a classic four stages transportation model in which the public transport

Table 7
MIS Latent Class models with different waiting time parameters.

	Waiting time parameter (α)			
	$\alpha = 0.5$	$\alpha = 0.75$	$\alpha = 1$	$\alpha = 1.5$
Disaggregated alternatives				
Travel time in bus ($\beta_{t_{bus}}$)	-0.056 (-28.2)	-0.061 (-29.7)	-0.085 (-42.7)	-0.086 (-42.2)
Travel time in Metro ($\beta_{t_{metro}}$)	-0.056 (-19.8)	-0.06 (-20.7)	-0.081 (-30.0)	-0.075 (-26.7)
Initial waiting time ($\beta_{t_{wt}}$)	-0.034 (-17.9)	-0.03 (-15.9)	-0.041 (-24.6)	-0.044 (-26.1)
Transfer waiting time ($\beta_{t_{wt}}$)	-0.037 (-7.5)	-0.035 (-7)	-0.043 (-9.0)	-0.031 (-6)
Transfer walking time ($\beta_{t_{wt}}$)	-0.136 (-13.1)	-0.174 (-16.3)	-0.396 (-39.6)	-0.4 (-39.2)
Bus crowding level (β_{CR})	-0.79 (-12.3)	-0.783 (-12)	-1.016 (-15.8)	-0.828 (-12.8)
PSC (β_{PSC})	-0.664 (-14.1)	-0.603 (-12)	-0.339 (-7.5)	-0.475 (-9.4)
Transfer bus-bus ($\beta_{t_{bb}}$)	-1.396 (-15.2)	-1.346 (-14.7)	-0.974 (-12.0)	-1.331 (-12.8)
Transfer bus-Metro ($\beta_{t_{bm}}$)	-0.849 (-12.5)	-0.866 (-12.5)	-0.938 (-14.0)	-0.872 (-11.9)
Transfer Metro-bus ($\beta_{t_{mb}}$)	-1.536 (-7.6)	-1.5 (-7.4)	-1.349 (-6.8)	-2.025 (-8.3)
Aggregated alternatives				
Travel time in bus ($\beta_{t_{bus}}$)	-0.075 (-28.5)	-0.065 (-24.6)	-0.05 (-19.1)	-0.035 (-13.4)
Travel time in Metro ($\beta_{t_{metro}}$)	-0.069 (-20.6)	-0.061 (-18)	-0.056 (-16.9)	-0.041 (-11.8)
Initial waiting time ($\beta_{t_{wt}}$)	-0.375 (-55.7)	-0.248 (-54.5)	-0.099 (-29.9)	-0.099 (-45.5)
Transfer waiting time ($\beta_{t_{wt}}$)	-0.144 (-7.8)	-0.092 (-7.3)	-0.058 (-6.3)	-0.044 (-7.6)
Transfer walking time ($\beta_{t_{wt}}$)	-0.528 (-35.9)	-0.485 (-33.4)	-0.08 (-5.8)	-0.24 (-17.3)
Bus crowding level (β_{CR})	0.174 (1.3)	0.132 (1)	-0.472 (-3.9)	-0.31 (-2.3)
PSC (β_{PSC})	-1.076 (-16.2)	-0.767 (-10.5)	0.189 (2.6)	-0.264 (-3.4)
Transfer bus-bus ($\beta_{t_{bb}}$)	-0.572 (-6.8)	-0.647 (-7.7)	-0.944 (-11.5)	-0.604 (-7.9)
Transfer bus-Metro ($\beta_{t_{bm}}$)	-0.806 (-11.4)	-0.746 (-10.6)	-0.722 (-12.1)	-0.532 (-8.3)
Transfer Metro-bus ($\beta_{t_{mb}}$)	-0.989 (-3.7)	-0.882 (-3.4)	-0.85 (-3.4)	-0.487 (-2.3)
Class Membership Model				
Constant disaggregated strategy (β_{cl})	-9.754 (-11.1)	-10.64 (-10.4)	-19.25 (-7.4)	-47.48 (-7.9)
Gini indicator (β_{γ})	24.239 (11.2)	26.044 (10.4)	50.22 (7.5)	113 (7.9)
Residual (β_{δ})	-47.673 (-12.5)	-52.819 (-11.4)	-108.3 (-7.4)	-160.9 (-7.8)
N° observations	150,430	150,430	150,430	150,430
Log-likelihood	-157,170.5	-157,332.8	-155,825.7	-157,998.8

assignment is affected by the car assignment, and the overall impact of the proposed changes is accounted for. The type and amount of data, as well as the computational burden needed to address such problem, exceeds the scope of the present paper.

Further research should focus on the revision of the assumption of the exponential distribution that is used for the waiting time. Although it has been found to fit real data well (Guo et al., 2011), recent efforts of improvement in this line have been explored in recent literature, including, among other things, mixing distributions (see, e.g. Ingvardson et al., 2018), and loglogistic, gamma, and erlang distributions (Li et al., 2015). It should be borne in mind that using those distributions may significantly complicate the problem by requiring, e.g., a complete enumeration of transfers and stops, although some approaches to that problem have been proposed in recent literature (Li et al., 2015).

Finally, the findings of our research can be used to suggest some guidelines for public transport models that can help steer decisions regarding where to implement new routes and how to improve existing ones. Firstly, given that a percentage of passengers use common lines, implementing lines with overlap in high-demand sectors will effectively allow passengers to reduce their waiting time. Secondly, as the model showed that bus-to-bus and Metro-to-bus transfers are more unpleasant than bus-to-Metro transfers, identifying OD pairs that required transfers would focus planning efforts to reduce the number of onerous transfers. Thirdly, given that our models show that transfers requiring walking generate larger disutility levels for passengers, high-demand transfer points should be carefully designed to avoid this problem.

CRediT authorship contribution statement

Jacqueline Arriagada: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Funding acquisition. **Marcela A. Munizaga:** Conceptualization, Methodology, Resources, Data curation, Writing – review & editing, Supervision, Funding acquisition. **Angelo Guevara:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Funding acquisition. **Carlo Prato:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

influence the work reported in this paper.

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Appendix

In this section, we tested the stability of the parameters of the MIS latent class model with respect to the alpha parameter, value of the waiting time parameter to define the common lines, considering that different values would result in different aggregated strategies. Specifically, we estimated 3 additional MIS latent class models with different alpha parameters (0.5, 0.75, and 1.5), that can be compared with the model estimated with alpha equal to 1.

In Table 7 we report the parameters of the 4 MIS latent class models estimated with different alpha parameters (0.5, 0.75, 1, and 1.5). In general, estimators for all models are statistically significant, with the expected sign. The only exceptions are the PSC term and the bus crowding level term. In all models the PSC is statistically significant however, only in the aggregated strategy class of the MIS Latent class model with $\alpha = 1$ the PSC term has the expected positive sign. On the other hand, for the aggregated strategy class, the bus crowding level term is statistically significant with the expected sign only in the models with $\alpha = 1$ and $\alpha = 1.5$. Both results suggest against using an alpha less than 1.

The results show some differences between models, however, they reproduce most results of this study, in particular, all models show the heterogeneous route choice strategies of passengers and that waiting time is more burdensome for passengers who use aggregated alternatives than passengers who use disaggregated alternatives. Besides, using an $\alpha = 1$ is more consistent in terms of expected parameters.

References

- Anderson, M.K., Nielsen, O.A., Prato, C.G., 2017. Multimodal route choice models of public transport passengers in the Greater Copenhagen Area. *EURO Journal on Transportation and Logistics* 6 (3), 221–245. <https://doi.org/10.1007/s13676-014-0063-3>.
- Arriagada, J., Gschwender, A., Munizaga, M.A., Trépanier, M., 2019. Modeling bus bunching using massive location and fare collection data. *Journal of Intelligent Transportation Systems: Technology, Planning, and Operations* 23 (4), 332–344. <https://doi.org/10.1080/15472450.2018.1494596>.
- Bagchi, M., White, P.R., 2005. The potential of public transport smart card data. *Transport Policy* 12 (5), 464–474. <https://doi.org/10.1016/j.tranpol.2005.06.008>.
- Bohara, A.K., Caplan, A.J., Grijalva, T., 2007. The effect of experience and quantity-based pricing on the valuation of a curbside recycling program. *Ecological Economics* 64 (2), 433–443. <https://doi.org/10.1016/j.ecolecon.2007.02.033>.
- Bovy, P.H.L., Bekhor, S., Prato, C.G., 2008. The Factor of Revisited Path Size. *Transportation Research Record: Journal of the Transportation Research Board* 2076 (1), 132–140. <https://doi.org/10.3141/2076-15>.
- Bovy, P.H.L., Hoogendoorn-Lanser, S., 2005. Modelling route choice behaviour in multi-modal transport networks. *Transportation* 32 (4), 341–368. <https://doi.org/10.1007/s11116-004-7963-2>.
- Chriqui, C., Robillard, P., 1975. Common Bus Lines. *Transportation Science* 9 (2), 115–121. <https://doi.org/10.1287/trsc.9.2.115>.
- Cominetti, R., Correa, J., 2001. Common-Lines and Passenger Assignment in Congested Transit Networks. *Transportation Science* 35 (3), 250–267. <https://doi.org/10.1287/trsc.35.3.250.10154>.
- Daly, A., Hess, S., de Jong, G., 2012. Calculating errors for measures derived from choice modelling estimates. *Transportation Research Part B: Methodological* 46 (2), 333–341. <https://doi.org/10.1016/j.trb.2011.10.008>.
- De Cea, J., Fernandez, J. E., Dekock, V., Soto, A., & Friesz, T. L. (2003). ESTRAUS: a computer package for solving supply-demand equilibrium problems on multimodal urban transportation networks with multiple user classes. In *In annual meeting of the Transportation Research Board, Washington, DC*.
- De Cea, J., Fernández, E., 1993. Transit Assignment for Congested Public Transport Systems: An Equilibrium Model. *Transportation Science* 27 (2), 133–147. <https://doi.org/10.1287/trsc.27.2.133>.
- De Grange, L., Raveau, S., González, F., 2012. A Fixed Point Route Choice Model for Transit Networks that Addresses Route Correlation. *Procedia - Social and Behavioral Sciences* 54, 1197–1204. <https://doi.org/10.1016/j.sbspro.2012.09.834>.
- Eluru, N., Chakour, V., El-Geneidy, A.M., 2012. Travel mode choice and transit route choice behavior in Montreal: Insights from McGill University members commute patterns. *Public Transport* 4 (2), 129–149. <https://doi.org/10.1007/s12469-012-0056-2>.
- Gallo, G., Pallottino, S., 1988. Shortest path algorithms. *Annals of Operations Research* 13 (1), 1–79. <https://doi.org/10.1007/BF02288320>.
- Glasser, Gerald J., 1962. *Variance Formulas for the Mean Difference and Coefficient of Concentration*. *Source*. *Journal of the American Statistical Association* 57 (299), 648–654.
- Godachevich, Javiera, Tirachini, Alejandro, 2020. Does the measured performance of bus operators depend on the index chosen to assess reliability in contracts? An analysis of bus headway variability. *Research in Transportation Economics* 101000. <https://doi.org/10.1016/j.retrec.2020.101000>.
- Greene, W.H., Hensher, D.A., 2003. A latent class model for discrete choice analysis: contrasts with mixed logit. *Transportation Research Part B: Methodological* 37 (8), 681–698.
- Grisson, E., Burkhardt, J.-M., Gyselinck, V., 2017. How do users choose their routes in public transport? The effect of individual profile and contextual factors. *Transportation Research Part F: Traffic Psychology and Behaviour* 51, 24–37. <https://doi.org/10.1016/j.trf.2017.08.011>.
- Gschwender, A., Munizaga, M., Simonetti, C., 2016. Using smart card and GPS data for policy and planning: The case of Transantiago. *Research in Transportation Economics* 59, 242–249. <https://doi.org/10.1016/j.retrec.2016.05.004>.
- Guevara, C.A., 2015. Critical assessment of five methods to correct for endogeneity in discrete-choice models. *Transportation Research Part A: Policy and Practice* 82, 240–254. <https://doi.org/10.1016/j.tra.2015.10.005>.
- Guevara, C.A., Polanco, D., 2016. Correcting for endogeneity due to omitted attributes in discrete-choice models: the multiple indicator solution. *Transportmetrica A: Transport Science* 12 (5), 458–478. <https://doi.org/10.1080/23249935.2016.1147504>.
- Guevara, C., Angelo, Tirachini, Alejandro, Hurtubia, Ricardo, Dekker, Thijs, 2018. Correcting for endogeneity due to omitted crowding in public transport choice using the Multiple Indicator Solution (MIS) method. *Transportation Research Part A: Policy and Practice*. 137, 472–484. <https://doi.org/10.1016/j.tra.2018.10.030>.

- Guo, S., Yu, L., Chen, X., Zhang, Y., 2011. Modelling waiting time for passengers transferring from rail to buses. *Transportation Planning and Technology* 34 (8), 795–809. <https://doi.org/10.1080/03081060.2011.613589>.
- Guo, Z., 2011. Mind the map! The impact of transit maps on path choice in public transit. *Transportation Research Part A: Policy and Practice* 45 (7), 625–639. <https://doi.org/10.1016/j.tra.2011.04.001>.
- Hoogendoorn-Lanser, S., Bovy, P., 2007. Modeling overlap in multimodal route choice by including trip part-specific path size factors. *Transportation Research Record* 2003 (1), 74–83.
- Hoogendoorn-Lanser, S., van Nes, R., Bovy, P., 2005. Path Size Modeling in Multimodal Route Choice Analysis. *Transportation Research Record: Journal of the Transportation Research Board* 1921 (1), 27–34. <https://doi.org/10.1177/0361198105192100104>.
- INRO (1996). EMME/2 User's Manual.
- Ingvarsson, J.B., Nielsen, O.A., Raveau, S., Nielsen, B.F., 2018. Passenger arrival and waiting time distributions dependent on train service frequency and station characteristics: A smart card data analysis. *Transportation Research Part C: Emerging Technologies* 90, 292–306. <https://doi.org/10.1016/j.trc.2018.03.006>.
- Jánošíková, L., Slavík, J., Koháni, M., 2014. Estimation of a route choice model for urban public transport using smart card data. *Transportation Planning and Technology* 37 (7), 638–648. <https://doi.org/10.1080/03081060.2014.935570>.
- Kim, I., Kim, H.-C., Seo, D.-J., Jung, J., & Kim, I. (2020). Calibration of a transit route choice model using revealed population data of smartcard in a multimodal transit network. *Transportation*, 47, 2179–2202. <https://doi.org/10.1007/s11116-019-10008-8>.
- Kusakabe, T., Iryo, T., Asakura, Y., 2010. Estimation method for railway passengers' train choice behavior with smart card transaction data. *Transportation* 37 (5), 731–749. <https://doi.org/10.1007/s11116-010-9290-0>.
- Li, Q., Chen, P., Nie, Y., 2015. Finding optimal hyperpaths in large transit networks with realistic headway distributions. *European Journal of Operational Research* 240 (1), 98–108. <https://doi.org/10.1016/j.ejor.2014.06.046>.
- Mcfadden, D. (2000). *Disaggregate Behavioral Travel Demand's RUM Side A 30-Year Retrospective*.
- Munizaga, M.A., Palma, C., 2012. Estimation of a disaggregate multimodal public transport Origin-Destination matrix from passive smartcard data from Santiago, Chile. *Transportation Research Part C: Emerging Technologies* 24, 9–18. <https://doi.org/10.1016/j.trc.2012.01.007>.
- Nassir, Neema, Hickman, Mark, Ma, Zhen-Liang, 2018. A strategy-based recursive path choice model for public transit smart card data. *Transportation Research Part B: Methodological* 126, 528–548. <https://doi.org/10.1016/j.trb.2018.01.002>.
- Nassir, N., Hickman, M., Ma, Z., Nassir, N., 2017. Statistical Inference of Transit Passenger Boarding Strategies from Farecard Data. *Transportation Research Record: Journal of the Transportation Research Board* 2652, 8–18. <https://doi.org/10.3141/2652-02>.
- Nguyen, S., Pallottino, S., 1988. Equilibrium traffic assignment for large scale transit networks. *European Journal of Operational Research* 37 (2), 176–186. [https://doi.org/10.1016/0377-2217\(88\)90327-X](https://doi.org/10.1016/0377-2217(88)90327-X).
- Nielsen, O.A., Eltvéd, M., Anderson, M.K., Prato, C.G., 2021. Relevance of detailed transfer attributes in multimodal route choice models for public transport passengers. *Transportation Research Part A: Policy and Practice*.
- Prato, C.G., 2009. Route choice modeling: past, present and future research directions. *Journal of Choice Modelling* 2 (1), 65–100. [https://doi.org/10.1016/S1755-5345\(13\)70005-8](https://doi.org/10.1016/S1755-5345(13)70005-8).
- Raveau, S., Muñoz, J.C., 2014. Analyzing route choice strategies on transit networks. Retrieved from. <https://trid.trb.org/view/1289224>.
- Raveau, Sebastián, Guo, Z., Muñoz, J.C., Wilson, N.H.M., 2014. A behavioural comparison of route choice on metro networks: Time, transfers, crowding, topology and socio-demographics. *Transportation Research Part A: Policy and Practice* 66 (1), 185–195. <https://doi.org/10.1016/j.tra.2014.05.010>.
- Raveau, Sebastián, Muñoz, J.C., de Grange, L., 2011. A topological route choice model for metro. *Transportation Research Part A: Policy and Practice* 45 (2), 138–147. <https://doi.org/10.1016/j.tra.2010.12.004>.
- Schmöcker, Jan-Dirk, Shimamoto, Hiroshi, Kurauchi, Fumitaka, 2013. Generation and calibration of transit hyperpaths q. 36, 406–418. <https://doi.org/10.1016/j.trc.2013.06.014>.
- Spies, H., Florian, M., 1989. Optimal strategies: A new assignment model for transit networks. *Transportation Research Part B* 23 (2), 83–102. [https://doi.org/10.1016/0191-2615\(89\)90034-9](https://doi.org/10.1016/0191-2615(89)90034-9).
- Swait, J., Ben-Akiva, M., 1987. Incorporating random constraints in discrete models of choice set generation. *Transportation Research Part B* 21 (2), 91–102. [https://doi.org/10.1016/0191-2615\(87\)90009-9](https://doi.org/10.1016/0191-2615(87)90009-9).
- Tan, R., Adnan, M., Lee, D.-H., Ben-Akiva, M.E., 2015. New Path Size Formulation in Path Size Logit for Route Choice Modeling in Public Transport Networks. *Transportation Research Record: Journal of the Transportation Research Board* 2538 (1), 11–18. <https://doi.org/10.3141/2538-02>.
- Van Der Hurk, E., Kroon, L., Maróti, G., Vervest, P., 2015. Deduction of passengers' route choices from smart card data. *IEEE Transactions on Intelligent Transportation Systems* 16 (1), 430–440. <https://doi.org/10.1109/TITS.2014.2333583>.
- Vrtic, M. J., & Axhausen, K. W. 2002. The impact of tilting trains in Switzerland: A route choice model of regional and long distance public transport trips paper submitted to the 82nd Annual Meeting of the Transportation Research Board. <https://doi.org/10.3929/ethz-a-004403563>.
- Walker, J.L., 2001. *Extended discrete choice models: integrated framework, flexible error structures, and latent variables*. Massachusetts Institute of Technology.
- Yap, M., Cats, O., van Arem, B., 2020. Crowding valuation in urban tram and bus transportation based on smart card data. *Transportmetrica A: Transport Science* 16 (1), 23–42. <https://doi.org/10.1080/23249935.2018.1537319>.
- Zhao, Juanjuan, Zhang, Fan, Tu, Lai, Xu, Chengzhong, Shen, Dayong, Tian, Chen, Li, Xiang-Yang, Li, Zhengxi, 2017. Estimation of Passenger Route Choice Pattern Using Smart Card Data for Complex Metro Systems. *IEEE Transactions on Intelligent Transportation Systems* 18 (4), 790–801. <https://doi.org/10.1109/TITS.2016.2587864>.